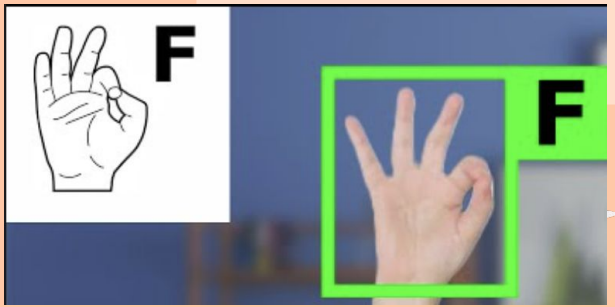
Silent Voice Bridge: The Edge AI ASL Interpreter



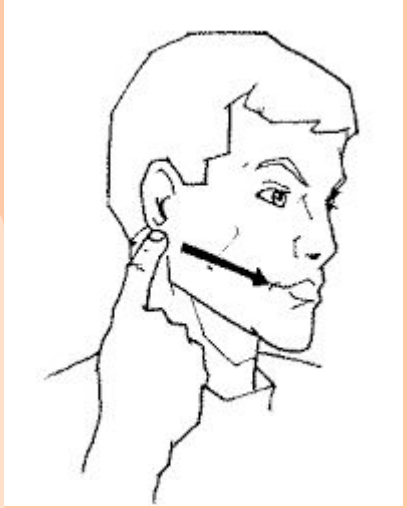
Autonomous Real-Time ASL Translation & Education

Presented By: Aramish Farooq, Nathaniel Ortega, Reuben George, and Pwint Thu Thu San



The Problem

- **Communication Gap:** 466 million people with disabling hearing loss vs. the hearing world.
- **Dependency:** Human interpreters are scarce; writing is slow.
- **Privacy**
- **Education:** Lack of real-time feedback hinders self-learning.





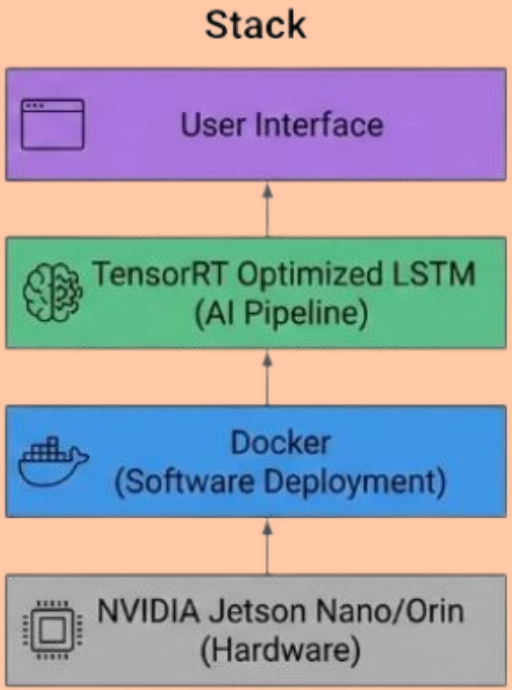
Why does It Matter?

- **Access is a Human Right**
  Communication is foundational to healthcare, education, employment, and safety.
- **Privacy is Non-Negotiable**
  Medical, legal, and financial conversations require confidentiality — cloud apps and third-party interpreters compromise this.
- **Independence Builds Dignity**
  No one should have to rely on another person for everyday communication.
- **Real Inclusion Requires Real-Time Solutions**
  Delayed or text-based communication limits spontaneity, emotion, and equal participation.



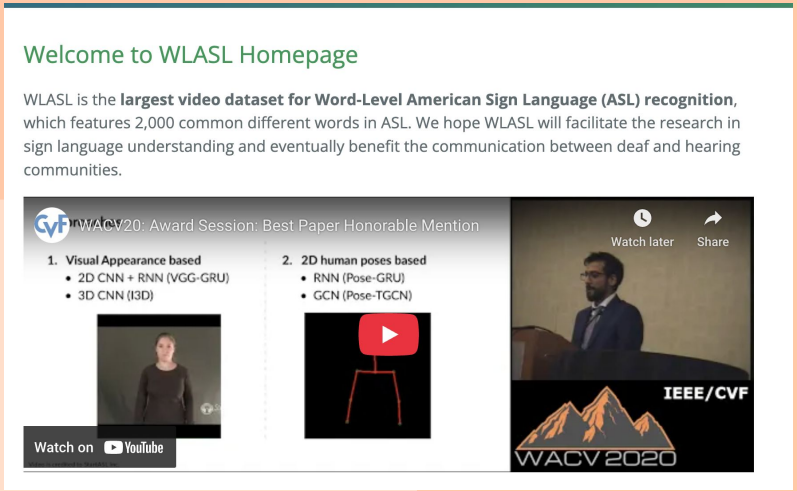
The Solution

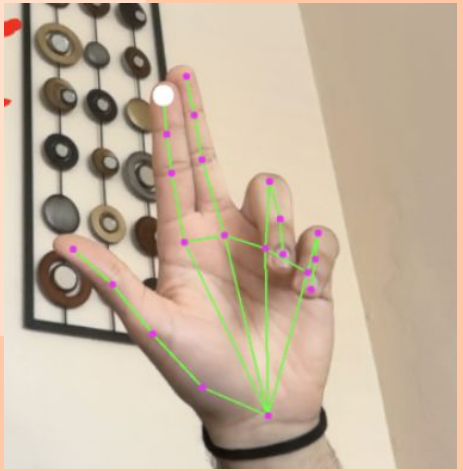
- **Approach:** Hardware/Software Co-Design.
- **Software Deployment**: Docker container for reproducible, plug-and-play updates. c).
- **Hardware:** NVIDIA Jetson Nano/Orin (1024 CUDA Cores).
- **AI Pipeline:** MediaPipe (Tracking) → **TensorRT** Optimized LSTM (Logic)



Feasibility

- **Foundation:** We use the industry-standard **WLASL Dataset** (2,000 signs) to teach the model general ASL.
- **Precision:** We fine-tune the model with **our own recordings** to ensure it works perfectly for our live demo.
- We will hold weekly meetings with the advisor.
- Showcase our work every week and updates

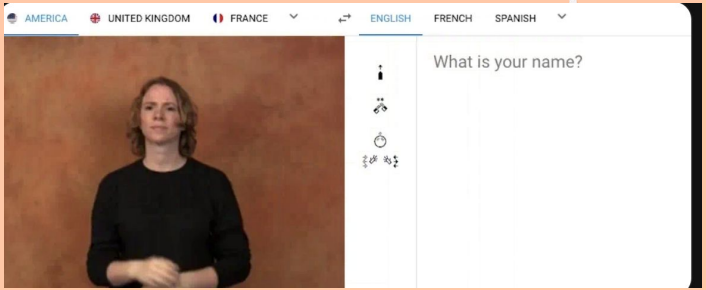




Conclusion

What We're Building:

- Portable device that translates ASL to text instantly
- Runs entirely offline, no internet needed
- Easy to update with plug-and-play software



Impact:

- Reduce dependence on human interpreters
- Learn ASL with instant feedback
- Bridging the gap for 466M+ people worldwide

